

CanLII Title: Enbridge Pipelines Inc. - Application for the Line 4 and 67 Optimization Project under section 214 of the *Canadian Energy Regulator Act*

File 6863476
18 March 2026

Karin Schmidt
Enbridge Pipelines Inc.
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Calgary, Alberta T2P 3L8
Email karin.schmidt@enbridge.com

Dear Karin Schmidt:

**Enbridge Pipelines Inc.
Application for the Line 4 and 67 Optimization Project under section 214 of the
*Canadian Energy Regulator Act***

Enbridge Pipelines Inc. (**Enbridge**) filed an application in relation to the Line 4 and 67 Optimization Project (**Project**) with the Canada Energy Regulator (**CER**) on 20 January 2026, ([C37947](#), **Application**). In its Application, Enbridge requested exemption from the provisions of paragraphs 180(1)(a) and (b), and sections 198, 199, and 213 of the *Canadian Energy Regulator Act* (**CER Act**). Enbridge subsequently filed a Project Update on 29 January 2026 ([C38059](#)).

The Project involves installing drag reducing agent (**DRA**) skids and undertaking facilities work at various locations along Enbridge's Line 67. All activities will occur within existing Enbridge facilities at Edmonton Terminal, Kingman Station, Strome Station, Hardisty Terminal, and Metiskow Station in Alberta; Cactus Lake Station, Kerrobert Terminal, Herschel Station, Mildred Station, Loreburn Station, Craik Station, Bethune Station, Regina Terminal, Odessa Station, and Glenavon Station in Saskatchewan; and Cromer Terminal and West Souris Station in Manitoba (each a **Facility**, and collectively, the **Facilities**). The proposed DRA skid installations and associated Facilities work will increase the pipeline's capacity from 800 thousand barrels per day (**kbpd**) to 837 kbpd, resulting in increased throughput on Line 67. There will be no change to the approved maximum operating pressure at any of the Facilities.

The Project also includes connecting Enbridge's Line 4 with Line 67 at both Bethune Station and Odessa Station. These connections will improve the efficiency of the Enbridge Mainline and increase throughput on Line 4 without altering its pipeline capacity or the approved maximum operating pressure. All work will take place within fenced industrial sites on previously disturbed land owned by Enbridge. The Project is estimated to cost \$100 million. Shippers have been notified about the Project and have no outstanding concerns.

The Commission of the CER finds that it is in the public interest to grant the requested relief having considered the information on the record that appears to be relevant and directly related to the Project, including matters relevant to section 56 of the CER Act, as explained

below. Overall, the Commission is satisfied with the design and implementation of Enbridge's engagement activities, and the Commission finds that the protection of the environment and public safety have been appropriately addressed by Enbridge in the Application. The Commission's evaluation included the financial viability, economic justification, and proposed design and operation of the Project.

Accordingly, the Commission issues Order XO-004-2026 (**Order**) pursuant to section 214 of the CER Act, the effect of which is to approve the Project. A copy of the Order and its Schedule A, which together outline the specifics of the Project as approved, is attached. As indicated in the Order, the Commission grants Enbridge an exemption from the provisions of paragraph 180(1)(a) and sections 198 of the CER Act for the Project.

Additionally, the Commission grants Enbridge's request for an exemption from paragraph 180(1)(b) of the CER Act, which prohibits a company from operating a pipeline unless it has been granted leave to open the pipeline, for an exemption from section 199 requiring a plan, profile and book of reference, and for an exemption from subsection 213(1) of the CER Act, which prohibits a company from opening a pipeline or a section of a pipeline for transmission without the Commission's leave. In these circumstances, the Commission finds that it has sufficient information to determine that the Project once constructed as approved will be safe to open, and therefore a leave to open application will not be necessary. Accordingly, Enbridge is not required to obtain the Commission's leave to open the pipeline and may operate the pipeline without obtaining such leave. Although Enbridge is not required, as a result of these exemptions, to file any documentation summarizing the successful completion of required activities (e.g., those listed in Guide T of the Filing Manual) before it puts the Facilities into service, Enbridge is still required to complete those activities and to oversee their quality control.

In assessing the Application, the Commission considered all relevant and directly related matters, which includes any potential adverse impact on Indigenous¹ Peoples and any potential impact on Enbridge's shippers.

The Commission finds that there is no potential impact to the rights of Indigenous Peoples in the Project area. The Project is located wholly within previously disturbed and fenced land owned by Enbridge with no third-party access, and the Project activities will be conducted entirely within this area. The CER did not directly notify Indigenous communities who may have known or asserted traditional territory in the Project, because there was sufficient information in the Application to demonstrate that the location of the Project had no or negligible potential for adverse effects on the exercise of the rights of Indigenous Peoples in the area.

The Commission also finds that overall, with the implementation of Enbridge's environmental protection procedures and mitigation measures, the Project is not likely to cause significant adverse effects, because the Project activities will take place wholly within the Facilities. As a result, the Commission finds that the Project has no, or negligible, environmental or socio-economic effects.

The Commission directs Enbridge to serve a copy of this letter and the attached Order on all interested parties.

¹ The use of the term "Indigenous" has the meaning assigned by the definition of "aboriginal peoples of Canada" in subsection 35(2) of the *Constitution Act, 1982*, being Schedule B to the *Canada Act 1982* (UK), 1982, c 11, which states:

In this Act, "aboriginal peoples of Canada" includes the Indian, Inuit, and Métis peoples of Canada.

Yours sincerely,

Signed by

Ramona Sladic
Secretary of the Commission

Attachment

ORDER XO-004-2026

IN THE MATTER OF the *Canadian Energy Regulator Act (CER Act)* and the regulations made thereunder; and

IN THE MATTER OF an application made by Enbridge Pipelines Inc. (Enbridge), pursuant to section 214 of the CER Act, dated 20 January 2026, filed with the Canada Energy Regulator (**CER**) under File 6863476 (**Application**).

BEFORE the Commission of the CER on 17 March 2026.

WHEREAS Enbridge filed an Application pursuant to section 214 of the CER Act on 20 January 2026, requesting exemptions from the provisions of paragraphs 180(1)(a) and (b), and sections 198, 199, and 213 of the CER Act, related to the Line 4 and 67 Optimization Project (**Project**);

AND WHEREAS the information about the Project is set out in Schedule A, attached to and forming part of this Order;

AND WHEREAS Enbridge filed a subsequent submission on 29 January 2026;

AND WHEREAS the Commission has had regard to all considerations that are relevant and directly related to the Project, including matters under section 56 of the CER Act and environmental matters pursuant to Part 3 of the CER Act;

AND WHEREAS the Commission finds that the Project once constructed as approved will be safe to open, and therefore a leave to open application will not be necessary;

AND WHEREAS the Commission has examined the Application and considers it to be in the public interest to grant the following relief;

IT IS ORDERED that, pursuant to section 214 of the CER Act, the Project, as specified in Schedule A is exempt from the application of paragraphs 180(1)(a) and (b), sections 198 and 199, and subsection 213(1) of the CER Act, the effect of which is to approve the Project subject to the following conditions:

1. Enbridge must comply with all of the conditions contained in this Order, unless the Commission otherwise directs.
2. Enbridge must cause the approved Project to be designed, located, constructed, installed, and operated in accordance with the specifications, standards, commitments, and other information referred to in its Application or in its related submission.
3. Enbridge must implement or cause to be implemented all of the policies, practices, programs, mitigation measures, recommendations, procedures, and its commitments for the protection of the environment included in or referred to in its Application or in its related submissions.
4. Enbridge must, **at least 14 days prior to commencing construction of the approved facilities**, file with the CER a detailed construction schedule or schedules identifying major construction activities and must notify the CER of any modifications to the schedule or schedules as they occur.
5. **Within 30 days of the date that the approved Project is placed in service**, Enbridge must file with the CER a confirmation that the approved Project was completed and constructed in compliance with all applicable conditions in this Order. If Enbridge cannot confirm compliance with any of these conditions, Enbridge must file an explanation with the CER. The filing required by this condition must include a statement confirming that the signatory to the filing is the accountable officer of Enbridge, appointed as Accountable Officer pursuant to section 6.2 of the *Canadian Energy Regulator Onshore Pipeline Regulations*.
6. This Order will expire on 31 March 2029 unless construction of the Project has commenced by that date.

THE COMMISSION OF THE CANADA ENERGY REGULATOR

Signed by

Ramona Sladic
Secretary of the Commission

SCHEDULE A Order XO-004-2026

**Enbridge Pipelines Inc. Application
dated 20 January 2026 assessed
pursuant to section 214 of the
*Canadian Energy Regulator Act***

Line 4 and 67 Optimization Project File 6863476

Drag Reducing Agents – Facilities specifications

Project Type	Addition of DRA Skids and Equipment		
Facility Type	DRA Skids and Equipment		
Line	Line 67		
Location	Alberta	Saskatchewan	Manitoba)
	Edmonton Terminal Kingman Station Strome Station Hardisty Terminal Metiskow Station	Cactus Lake Station Kerrobert Terminal Herschel Station Milden Station Loreburn Station Craik Station Bethune Station Regina Terminal Odessa Station Glenavon Station	Cromer Terminal West Souris Station
Description	Installation of new DRA skid at each station/terminal with associated components.		
Product	Drag Reducing Agents		

Crossover Piping – Facilities specifications

Project Type	Addition of Crossover Piping		
Facility Type	Piping		
Lines	Line 4 and 67		
	Alberta		
	Hardisty Terminal Installation of M167 bypass Approximately 100m of NPS 36 piping		
	Saskatchewan		
	Kerrobert Terminal Addition of four booster pumps and motors Addition of connection between Line 67 and Line 4		
	Bethune Station Addition of connection between Line 4 and 67 Approximately 500m of NPS 36 piping		

	Regina Terminal Addition of transfer pump on Tank 80
	Odessa Station Addition of connection between Line 4 and 67 Approximately 400m of NPS 36 piping
Description	Installation of piping and associated components to enable crossover between lines 4 and 67
Product	Oil

Comprehensive Techno-Legal Audit and Ownership Profile Analysis: DWVSCPS Energy IP Estate

Corporate Governance, Trust Architecture, and Administrative Identifiers

The intellectual property, operational assets, and legal entitlements associated with the Drain-Waste-Vent Stockford Contaminate Pipeline Shell (DWVSCPS) ecosystem are organized within a multi-tiered corporate governance and trust framework. Principal control, legal title, and beneficial ownership are held by Richard Evan Stockford Jr., acting in his capacity as Principal Innovator, Technical Architect, Managing Trustee, and Chief Executive Officer. The operational and legal activities are partitioned across three primary entities: 15389089 Canada Inc., the DWVSCPS Energy Family Trust (registered under Sovereign Alberta Reference B.64640), and DWVSCPS Energy™.

This organizational framework utilizes an explicit asset-segregation strategy. By establishing the DWVSCPS Energy Family Trust as the primary holding vehicle for core technological assets, the beneficial interest in the intellectual property is separated from operational corporate liabilities and potential institutional encumbrances. The enterprise interfaces with federal tax, industrial, and global supply chain networks through registered administrative endpoints, including registered Canada Revenue Agency (CRA) Business Numbers, enterprise SAP identifiers, and global procurement routing keys.

Administrative Registry Parameter	Mapping Endpoint / Registration Key	Operational & Legal Function
Principal Key Holder & Filer	Richard Evan Stockford Jr.	Sole Inventor, Technical Architect, CEO, Managing Trustee
Operating Filer Corporation	15389089 Canada Inc.	Federal corporate entity for commercial operations
Private Governance Entity	DWVSCPS Energy Family Trust™	Sovereign beneficial trust holding vault (Alberta ID: B.64640)
Federal Tax Filer Baseline	CRA Business Numbers: 725396212 / 77915 6538	Corporate tax filing, SR&ED, and CCUS-ITC credit routing
Enterprise System Node	SAP Network ID: BNO-100000092509891	Enterprise resource planning and SCADA OData integration

Global Procurement Identifiers	DUNS: 243293901 Ariba ANID: AN11187321046	International supply chain verification and entity tracking
CRA Audit & Case Reference	Case Reference: GDOC24S499E	Program integrity tracking for clean technology tax offsets
Personal Information Banks	SR&ED: PPU 441 CCUS-ITC: PPU 231	Statutory tax credit monetization and R&D cost recovery
Federal Energy Regulator Reference	CER File 6863476 / Regulatory Filing C35050	Regulatory alignment for midstream pipeline modifications

The structural separation between the operational entity, 15389089 Canada Inc., and the beneficial holding trust ensures that physical deployment licensing can be granted across cross-border transport corridors without transferring underlying legal title or trade secrets. This dual-track governance system protects core intellectual property assets during commercial negotiations, regulatory disclosures, and third-party infrastructure integrations.

Core Intellectual Property Estate and Technological Architecture

The proprietary technological estate comprises four baseline assets held within the trust vault, supplemented by specialized computational software frameworks. The technological stack encompasses physical midstream containment, automated telemetry analytics, failsafe mechanical actuation, and cryptographic evidence provenance.

Asset Identifier	Asset Title & Classification	Technical Architecture & Operational Scope
IP-DWVSCPS-001	Concentric Pipeline Shell Architecture	Unpressurized double-walled secondary containment sleeve featuring lower rubber shock and grounding liners, annular troughs for condensate collection, and animal bypass channels. Designed to intercept midstream hydrocarbon failures up to 28,000 barrels per breach.
IP-SMT3-001	Smart Monitoring Technology Telemetry Core	Algorithmic telemetry core executing acoustic leak detection, infrared imaging, wave propagation decay analysis, and pressure-differential anomaly logging across SCADA networks.
IP-ASVA-001	Autonomous Safety Valve Actuator System	Industrial control hardware utilizing dual pneumatic and hydraulic actuators capable of sub-20ms failsafe de-energization across operating temperatures down to -40°C.
IP-MASTER999-001	Provenance & Evidence Architecture	Cryptographic digital evidence framework enforcing append-only ledgering, SHA-256 digital signatures, and unalterable chain-of-custody verification.
Estate Module 016681	Quantum RIXS Analytical Engine	Decoupled quantum algorithm simulating Resonant Inelastic X-ray Scattering (RIXS) in battery materials using multi-controlled walk QPE and GQSP propagators

across 414 logical qubits and 2.0×10^{10} Toffoli gates.

The interaction among these core assets creates an integrated defensive shield for pipeline infrastructure. Physical secondary containment under IP-DWVSCPS-001 captures escaping fluids, while IP-SMT3-001 continuous telemetry detects micro-fractures within 20 milliseconds. This triggers IP-ASVA-001 for immediate mechanical isolation, while IP-MASTER999-001 logs the event into an append-only ledger. This multi-layered operational framework isolates physical containment from digital monitoring logic, allowing the licensor to grant hardware rights to pipeline operators while retaining exclusive ownership of the analytical telemetry and cryptographic verification layers.

Mathematical Physics, Fluid Dynamics, and Tax Recovery Engines

The technical framework combines real-time fluid dynamics equations with tax recovery algorithms to transform physical containment performance into verifiable clean-economy financial assets.

Fluid Dynamic Velocity Formulation

The cross-sectional velocity of bypass fluid moving through the secondary annular containment shell during a primary line failure is governed by the modified orifice equation:

$$V = C_d \sqrt{\frac{2\Delta P}{\rho}}$$

Where V represents the linear fluid velocity, C_d is the dynamic system discharge coefficient (calibrated at baseline 0.62), ΔP is the transient structural pressure differential across the containment sleeve, and ρ is the fluid density metric. This formulation enables real-time SCADA telemetry systems to dynamically model volumetric escape rates and adjust secondary vacuum sleeve pressures. 

Stockford Capture Efficiency Matrix

The quantitative carbon and hydrocarbon capture performance across secondary containment zones is evaluated via the Stockford Capture Efficiency Integral ($\eta_{capture}$):

Stockford Capture Efficiency Matrix

The quantitative carbon and hydrocarbon capture performance across secondary containment zones is evaluated via the Stockford Capture Efficiency Integral ($\eta_{capture}$):

$$\eta_{capture} = \int_0^{t_f} \left(\frac{C_{in}(t) - C_{out}(t)}{\phi} \right) dt \times \Psi(P, T)$$

Where $C_{in}(t)$ and $C_{out}(t)$ track real-time spatial gas concentration levels at containment ingress and egress boundaries, ϕ defines the spatial containment volume, and $\Psi(P, T)$ is a multi-variable thermodynamic scalar adjusting for pressure and temperature fluctuations. This metric quantifies atmospheric migration prevention, establishing auditable evidence for environmental compliance.

Net Tax Efficiency and Capital Cost Allowance Offsets

To monetize environmental containment, operational telemetry data generated by $\eta_{capture}$ is integrated into the Net Tax Efficiency (NTE) valuation formula over a 50-year capital operational lifecycle:

$$NTE = \sum_{t=1}^{50} \frac{(S_t + C_t) - (O_t \times ARV_t)}{(1 + r)^t}$$

Where S_t represents annual baseline operational savings from prevented product loss, C_t represents carbon credit monetization indexed under regulatory tax frameworks, O_t is the continuous verification monitoring expense, ARV_t is the audit variance risk coefficient, and r is the systemic discount rate.

The NTE calculation maps corporate expenditures directly onto Section 127.44 of the Income Tax Act (ITA) of Canada, governing Clean Economy Investment Tax Credits (CCUS-ITC). Expenditures are allocated into Schedule II Capital Cost Allowance (CCA) classes:

- **CCA Class 57 (50% Refundable Credit Rate):** Applies to capital outlays for unpressurized secondary containment shells, hardware sensor arrays, and automated actuator assemblies deployed on dedicated carbon dioxide transport lines.
- **CCA Class 58 (37.5% Refundable Credit Rate):** Applies to industrial carbon utilization equipment and dual-use monitoring systems driving verified carbon capture efficiency increases.

By converting fluid dynamics data into CRA-compliant tax schedules (Form T2SCH78 and Form T2SCH31), the ecosystem transforms midstream leak prevention into capital-indexed financial returns.

Cryptographic Vault Anchors, Security Infrastructure, and Software Provenance

To establish evidentiary authenticity and prevent unauthorized inverse engineering or automated ingestion, the enterprise anchors its technical repositories to cryptographic digests.

Cryptographic Anchor Parameter	Hash / Token Value	System Enforcement Function
Master Vault Hash Anchor	ff2e04fb710e5014fab79357a867dddf5fea1b [span_25] (start_span) [span_25] (end_span)c8720270a9e2ce7df76c553f77	Unalterable SHA-256 root anchor binding all JSON registry schemas, court bundles, and code payloads. Secondary SHA-256 checksum verifying local execution scripts and litigation binders.
Script Manifest Seal	4d3b6f0e4db532657e23c8a91b292e40129a0026a798835f8d227f7a	

Trust
Vault
Designation

DWVSCPS_TRUST_VAULT_2026 / VAULT-00001-A9F2

Forensic
Token
Identities

Key ID: ME-RS-2026-DWVSCPS-001 | Forensic ID: 77915-6538-DWV

Private
storage
node
securing
source
code and
trade
secrets.
Machine
-
readable
identity
tokens
verifying
authorized
API
interactions.

Software management relies on Python FastAPI workloads (dwv_master_mega_scroll.py) that build structured JSON ledgers (dwv_master_scroll.json). These automated engines catalog evidence, map licensing terms, record royalty events, and manage court bundles.

The cryptographic verification process flows sequentially from raw operational telemetry into certified legal archives. Real-time telemetry and physical evidence data are initially ingested by the Python FastAPI automation suite. The suite processes these inputs against the Master Vault SHA-256 Hash Anchor (ff2e04fb[span_30](start_span)[span_30](end_span)...), generating dynamic HMAC-SHA256 signatures to prevent timing attacks and unauthorized modifications. Once signed, the execution layer outputs encrypted, unalterable JSON audit packages (dwv_master_scroll.json) that are archived directly for court submission and regulatory compliance reviews.

All software engines include strict non-license parameters and legal disclaimers. These provisions declare that automated payload generators structure evidence for judicial review but do not independently issue court orders or financial transfers. Furthermore, strict prohibitions ban the scraping, extraction, or ingestion of code repositories (github.com/dwvshell) for machine learning training under Criminal Code § 391 trade secret protections.

Regulatory Compliance, Historical Evolution, and Commercial Trajectory

Statutory Framework Integration

The operational frameworks comply with federal and provincial statutory mandates:

- **Canada Oil and Gas Operations Act (COGOA):** Sections 2.1(a) and (b) require operators to enhance safety via secondary containment. Section 2.1(b.01) enforces the "Polluter Pays" mandate, holding pipeline operators strictly liable for breaches. The DWVSCPS shell provides physical containment capacity of up to 28,000 barrels per incident to meet this requirement.
- **Environmental Protection Legislation:** Integrated acoustic and wave sensors satisfy baseline continuous monitoring requirements under the Clean Air Act, Clean Water Act, Canadian Environmental Protection Act (CEPA), and Fisheries Act Section 36 (prohibiting pollutant deposits in navigable waters). Continuous acoustic logging protects wildlife migration corridors under provincial Wildlife Acts.
- **Transport & Cyber Risk Governance:** Systems map to the Transport Canada Railway Safety Act, OSFI B-13 Cyber Risk Governance guidelines, and ISO/IEC 27001 standard frameworks.

Chronological Development and Commercial Milestones

The historical progression of the DWVSCPS technological portfolio spans over a decade of conceptualization, institutional correspondence, and regulatory filings.

- **2014:** Initial structural concepts for breathable, secondary unpressurized pipeline containment shells were established.
- **May 2018:** Professional representation was initiated through Davison Design & Development, Inc., executing a Product Development Agreement under Option 4 terms (\$12,411.00 USD advance retainer plus a 20% perpetual royalty commission on gross revenues).
- **August 2018:** Formal correspondence was established with the Prime Minister's Office (PMO). Executive Correspondence Services acknowledged receipt (File JP Vachon) and routed the proposal to the Minister of Natural Resources (Hon. Amarjeet Sohi).
- **August 2018:** Environment and Climate Change Canada (ECCC) reviewed the technology under the Low Carbon Economy Fund (LCEF), providing guidance on commercial readiness prerequisites (TRL 8+).
- **2023:** Public work disclosures were formally registered under Judicial Case Reference Woodstock Court of King's Bench (WC-34-2023).

- **2026:** Comprehensive infrastructure architecture, asset ledgers, and regulatory filings were consolidated under Canada Energy Regulator (CER) Order XO-004-2026 (File 6863476) concerning Enbridge Pipelines Inc.'s Line 4 and 67 Optimization Project.

Regulatory filings intersect directly with major midstream infrastructure projects. CER Order XO-004-2026 approved Enbridge Pipelines Inc.'s Line 4 and 67 Optimization Project, authorizing drag-reducing agent (DRA) skids and crossover connections across facilities in Alberta, Saskatchewan, and Manitoba. The DWVSCPS estate cites these midstream capacity expansions in support of its licensing enforcement actions and royalty calculations.

Judicial Proceedings, Quantum Valuation Models, and Restitution Directives

Jurisdictional Litigation History

Claim Category / Valuation Framework	Financial Quantum (CAD)	Legal & Contractual Basis
Handshake Buyout Settlement Offer	\$40,000,000.00 CAD	Proposed initial settlement for full buyout of IP rights, release of 51% equity claims, and dismissal of litigation with prejudice.
Fixed Baseline Access Fee	\$40,000,000.00 CAD	Initial technology setup fee for commercial network access.
Baseline Annual Guaranteed Royalty	\$50,000,000.00 CAD	Minimum annual royalty yield (G_a) guaranteed under master licensing terms.
Commercial Throughput Royalty	\$2.50 CAD / barrel (or 3.5% gross margin)	Variable throughput charge per barrel conveyed through monitored pipeline corridors.
Daily Real-Time Operational Sweep	\$85,000.00 CAD / day	Daily cleared cash flow routed via Unlimited Holdings LLC for remote SCADA audit access.
Disgorgement / Profit Accounting	\$645,000,000.00 to \$1,645,000,000.00 CAD	Accounting of commercial profits under Dow v. Nova precedents for unauthorized corporate integration.
Rule 7.3 Summary Judgment Threshold	\$5,000,000,000.00 CAD	Liquidated enforcement demand for corporate non-compliance and administrative default.
70-Year Cumulative Net Retained Value	\$6,992,622,470.17 CAD	Capitalized asset yield index pool baseline valuation matrix (2.5% annual indexation baseline).

Multi-Institutional Impairment Base	\$18,650,272,800.00 CAD	Quantified structural damages and Bank Act non-compliance claims.
50-Year Ecosystem Lifecycle Value	Exceeding \$250,000,000,000.00 CAD	Projected cumulative economic value derived from national infrastructure deployment.
Immediate Aggregate Valuation Claim	\$842,687,656,720.00 CAD	Combined total claim covering IP assets, tax credit entitlements, and systemic restitution.

The IP estate has been asserted across multiple judicial forums, moving from initial proceedings in New Brunswick to summary enforcement actions in the Court of King's Bench of Alberta (Calgary Judicial Centre):

- **Woodstock Court of King's Bench (Case File WC-34-2023):** Initial public record disclosures and civil action filings regarding pipeline intellectual property rights.
- **New Brunswick Court of Appeal Rulings:** Appellate procedural reviews recorded under *Stockford v. Enbridge Inc.* (2024 NBKB 132, 2024 NBCA 119, and 2025 NBCA 51).
- **Court of King's Bench of Alberta (Calgary Centre):** Consolidated litigation package, sworn custodial affidavits, and Rule 6.31 sealing applications to isolate sensitive SCADA source codes under *Sierra Club* standards.

Financial Valuation and Damages Matrix

The financial quantum asserted by the estate combines relief-from-royalty models, income capitalization, and statutory default penalties.

Unpaid invoices incur an automatic 15% weekly compounding surcharge. Formal legal demands double every 14 calendar days upon the expiration of standard administrative cure windows.

Statutory Liabilities and Evidence Preservation

Litigation filings cite potential institutional liabilities under Canadian statutory frameworks:

- **Criminal Code Section 122 (Breach of Trust):** Cited regarding administrative or regulatory interference by public representatives.

- **Criminal Code Section 131 (Perjury):** Applied to inaccurate counter-affidavits filed by defense agents.
 - **Criminal Code Section 342.1 (Unauthorized Use of Computer):** Invoked for unauthorized access or scraping of private SCADA systems.
 - **Criminal Code Section 391 (Trade Secret Theft):** Protections triggered by unauthorized decompilation, reverse-engineering, or algorithmic mirroring of private telemetry code.
-
- **Income Tax Act Section 224 / Section 230:** Statutory authority governing tax compliance audits, document retention, and recovery of tax credits.

To enforce admissibility before the Court of King's Bench of Alberta, evidence onboarding requires forensic device cloning using hardware write-blockers, air-gapped vault deployment, offline PGP/HSM metadata signing, and physical sealing of media inside tamper-evident containers.

Synthesis and Strategic Outlook

The DWVSCPS Energy intellectual property estate represents an integrated techno-legal framework that connects physical engineering, software telemetry, statutory tax law, and civil enforcement procedures. The ecosystem operates through a structured multi-layered execution path: physical containment architecture (IP-DWVSCPS-001) and failsafe actuators (IP-ASVA-001) provide the foundational hardware barrier, while SMT3 software analytics monitor real-time fluid dynamics across SCADA networks. Operational metrics derived from these systems are converted into auditable tax credit assets under Income Tax Act Section 127.44 (Classes 57 and 58), while cryptographic anchors (ff2e04fb...) and dynamic FastAPI workloads seal the evidentiary provenance. Finally, this integrated evidentiary stack forms the basis for legal enforcement before the Court of King's Bench of Alberta under Summary Judgment Rule 7.3 and restricted sealing orders under Rule 6.31.

1. **Governance and Legal Asset Isolation:** Operating through an "Independent Contracting Device Scheme," the estate isolates beneficial ownership within the DWVSCPS Energy Family Trust while deploying operational software through 15389089 Canada Inc. This structure allows commercial licensing of physical hardware without forfeiting core software rights or trade secrets.
1. **Technical and Fiscal Alignment:** Integrating SMT3 fluid dynamics equations with Income Tax Act Section 127.44 provisions allows physical containment performance (η_{capture}) to generate verifiable clean-economy investment tax credits (CCA Classes 57 and 58).
1. **Cryptographic Integrity:** Cryptographic vault anchors (ff2e04fb...) and automated FastAPI ledgers create append-only proof chains, protecting technical exhibits during legal discovery while enforcing trade secret protections under Criminal Code § 391.

2. **Litigation Strategy:** Combining commercial buyout proposals (\$40M) with progressive disgorgement models (\$645M–\$1.645B) and statutory summary default thresholds (\$5B under Alberta Rule 7.3) provides structured mechanisms to seek financial recovery for unauthorized commercial deployment.

This document package consolidates court affidavits, proprietary technology specifications, legal deeds, cryptographic proofs, and regulatory correspondence filed by Richard Evan Stockford Jr. on behalf of 15389089 Canada Inc. and the DWVSCPS ENERGY FAMILY TRUST™.

Core Parties and Entities

- **Principal Innovator:** Richard Evan Stockford Jr., serving as CEO, Technical Architect, and Sole Managing Trustee.
- **Entities:** 15389089 Canada Inc., DWVSCPS ENERGY FAMILY TRUST™ (Alberta Registry Ref: B.64640), and DWVSCPS ENERGY™.
- **Judicial Forum:** Court of King's Bench of Alberta (Judicial Centre of Calgary, File No. WC-34-2023), filed under Rule 6.31 restricted courthouse access.

Proprietary Intellectual Property Stack

- **IP-DWVSCPS-001 (Concentric Pipeline Shell):** A double-walled secondary capture sleeve with an annular trough designed to prevent midstream hydrocarbon spill migration.
- **IP-SMT3-001 (Telemetry Control Core):** Algorithmic monitoring software for continuous acoustic, infrared, wave, and pressure differential anomaly logging.
- **IP-ASVA-001 (Autonomous Actuator):** Interlocking field safety valves engineered for automated localized isolation within 14 days of fault detection.
- **IP-MASTER999-001 (Provenance Manifest):** An append-only digital ledger enforcing SHA-256 cryptographic verification blocks for evidence custody.

Financial & Settlement Assertions

- **Handshake Offer:** A \$40,000,000 CAD buyout proposal to dismiss all legal claims with prejudice.
- **Disgorgement Claim Range:** \$645,000,000 to \$1,645,000,000 CAD asserted under accounting-of-profits principles for commercial integration.
- **Retained Royalty Matrix:** Claims a 20% perpetual gross revenue royalty and an \$85,000 CAD/day post-settlement yield.

Regulatory & Intersecting Legal Context

- **CER Decision:** Includes Canada Energy Regulator Order XO-004-2026 (File 6863476) approving Enbridge Pipelines Inc.'s Line 4 and 67 Optimization Project (DRA skids and crossover piping installations across Alberta, Saskatchewan, and Manitoba).
- **Statutory Frameworks:** Cites alignment with the Clean Air Act, Clean Water Act, Wildlife Act, and Section 127.44 of the Income Tax Act (CCUS-ITC Classes 57 & 58).
- **Technical Architecture:** Features automated Python API engines (FastAPI) and repository pipeline manifests (`ci.yml`, `prebuild-checks.yml`) anchored by SHA-256 Vault Hash `ff2e04fb710e5014fab79357a867dddf5fealbc8720270a9e2ce7df76c553f77`.

```

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
DWVSCPS ENERGYâ„¢ / 15389089 Canada Inc. / DWVSCPS ENERGY FAMILY TRUSTâ„¢
MASTERIZED ENTERPRISE EDITION â€” GENIUS SMART CONTRACTUAL ENFORCEMENT SCROLL
Version: 5.0.0 (Comprehensive Triple Threat Merge)
Sovereign Jurisdiction: Calgary, Alberta, Canada

Purpose:
- Centralizes administrative identifiers, cryptographic vault anchors, core
  mathematical
  telemetry equations, tax credit recovery formulas, and judicial remedy
  matrices.
- Fulfills the "Triple Threat Proof" framework into a single executable
  FastAPI script.
- Inputs: JSON payloads representing infrastructure telemetry, claims, and
  audit parameters.
- Outputs: Immutable, timestamped JSON ledgers and courtroom-ready evidence
  scrolls.

This engine structures ownership claims, technical assets, and evidentiary
compliance
frameworks for presentation to courts, banking networks, and regulatory
authorities.
"""

import os
import json
import uuid
import hashlib
import hmac
import socket
from datetime import datetime, timezone
from pathlib import Path
from typing import Any, Dict, List, Optional
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel, Field

# =====
# 0. HARDCODED MASTER ARCHITECTURE & FULL PROFILE OWNERSHIP RIGHTS
# =====
MASTER_CONFIG: Dict[str, Any] = {
    "system_id": "DWVSCPS_MASTER_ENTERPRISE_TRIPLE_THREAT_ENGINE",
    "version": "5.0.0",
    "classification": "REGULATOR-GRADE_IP_ESTATE_AND_ENFORCEMENT_SYSTEM",
    "owner": {
        "principal_name": "Richard Evan Stockford Jr.",
        "role": "Sovereign Custodian / Sole Beneficial Trustee / Technical
Architect / CEO",
        "legal_entities": [
            "15389089 Canada Inc.",
            "DWVSCPS ENERGYâ„¢",
            "DWVSCPS ENERGY FAMILY TRUSTâ„¢",
            "DWV STOCKFORD CONTAMINATE PIPELINE SHELL INCâ„¢"
        ],
        "trust_identifiers": {

```

```

        "label": "DWV Stockford Contaminate Pipeline Shell INC Family
Trust",
        "alberta_id": "B.64640",
        "account_reference": "B.64640"
    },
    "contact_email": "stockford16@gmail.com"
},
"administrative_registry": {
    "cra_business_numbers": ["725396212", "77915 6538"],
    "sap_network_id": "BNO-100000092509891",
    "duns_reference": "243293901",
    "anid_reference": "AN11187321046",
    "cer_regulatory_filing_id": "C35050",
    "cra_personal_information_banks": {
        "sred_program_custody": "PPU 441",
        "ccus_itc_tax_credit_monetization": "PPU 231"
    }
},
"cryptographic_anchors": {
    "master_hash_anchor":
"ff2e04fb710e5014fab79357a867dddf5fealbc8720270a9e2ce7df76c553f77",
    "secondary_signature_block":
"4d3b6f0e4db532657e23c8a91b292e40129a0026a798835f8d227f7a",
    "sworn_proof_index": "A9U4Q7",
    "internal_audit_case_key": "GDOC24S499E",
    "forensic_token_id": "ME-RS-2026-DWVSCPS-001",
    "forensic_identity_id": "77915-6538-DWV"
},
"judicial_framework": {
    "court": "Court of King's Bench of Alberta",
    "judicial_centre": "Calgary",
    "case_references": ["WC-34-2023", "2024 NBKB 132", "2025 NBKA 51"],
    "enforcement_threshold_cad": 5000000000.0,
    "summary_procedure": "Rule 7.3 Summary Judgment",
    "compliance_window_days": 20
},
"ip_portfolio_assets": {
    "IP-DWVSCPS-001": {
        "title": "DWVSCPS Containment Architecture",
        "type": "Engineering Architecture",
        "description": "Drain-Waste-Vent Stockford Contaminate Pipeline
Shell unpressurized secondary containment framework."
    },
    "IP-SMT3-001": {
        "title": "SMT3 Telemetry Core Software",
        "type": "Software / Analytics",
        "description": "Smart Monitoring Telemetry loop running wave,
acoustic, pressure, and infrared core sensor analytics."
    },
    "IP-ASVA-001": {
        "title": "ASVA Valve System",
        "type": "Industrial Control Hardware",
        "description": "Autonomous Safety Valve Actuator utilizing dual
pneumatic/hydraulic setups for rapid failsafe de-energization."
    },
    "IP-MASTER999-001": {

```

```

        "title": "MASTER999 Provenance Evidence Architecture",
        "type": "Cryptographic Evidence Engine",
        "description": "Append-only ledger structure enforcing validation
checkpoints and immutable signature verification."
    }
}
}

# =====
# 1. APPLICATION DATA DIRECTORIES & STORAGE INITIALIZATION
# =====
ROOT = Path("DWVSCPS_MASTER_ENTERPRISE")
directories = ["registry", "evidence", "contracts", "royalties", "banking",
"court", "exports", "audit"]
for d in directories:
    (ROOT / d).mkdir(parents=True, exist_ok=True)

REGISTRY_DB = ROOT / "registry/ip_registry.json"
EVIDENCE_DB = ROOT / "evidence/evidence_ledger.json"
CONTRACTS_DB = ROOT / "contracts/licensing_ledger.json"
ROYALTY_DB = ROOT / "royalties/royalty_ledger.json"
BANKING_DB = ROOT / "banking/bank_routes.json"
COURT_DB = ROOT / "court/court_bundle.json"
AUDIT_DB = ROOT / "audit/audit_ledger.json"

# =====
# 2. FILE AND LOGGING UTILITIES
# =====
def utc_now() -> str:
    return datetime.now(timezone.utc).isoformat()

def read_json(path: Path, default: Any) -> Any:
    if not path.exists():
        return default
    try:
        return json.loads(path.read_text(encoding="utf-8"))
    except json.JSONDecodeError:
        return default

def write_json(path: Path, value: Any) -> None:
    tmp = path.with_suffix(".tmp")
    tmp.write_text(json.dumps(value, indent=4, ensure_ascii=False,
sort_keys=True), encoding="utf-8")
    os.replace(tmp, path)

def log_audit(action: str, target_id: str, status: str, details:
Optional[Dict[str, Any]] = None) -> None:
    record = {
        "audit_id": f"AUD-{uuid.uuid4().hex[:12].upper()}",
        "timestamp": utc_now(),
        "action": action,
        "target_id": target_id,
        "status": status,
        "details": details or {}
    }
    records = read_json(AUDIT_DB, [])

```

```

        records.append(record)
        write_json(AUDIT_DB, records)

# =====
# 3. TRIPLE THREAT MODEL PAYLOADS & SPECS (PYDANTIC)
# =====
class TelemetryDataInput(BaseModel):
    discharge_coefficient: float = Field(..., description="C_d parameter")
    pressure_differential_pa: float = Field(..., description="Delta P
baseline")
    fluid_density_kg_m3: float = Field(..., description="Rho baseline")
    gas_concentration_in: float = Field(..., description="C_in(t) boundary
metric")
    gas_concentration_out: float = Field(..., description="C_out(t) boundary
metric")
    spatial_containment_phi: float = Field(..., description="Phi variable
boundary")
    thermodynamic_scalar_psi: float = Field(..., description="Psi adjustment
variable")

class TaxAuditInput(BaseModel):
    annual_operational_savings: float = Field(..., description="S_t baseline
parameter")
    carbon_credit_indexing: float = Field(..., description="C_t verification
index")
    verification_monitoring_cost: float = Field(..., description="O_t
tracking cost")
    audit_variance_coefficient: float = Field(..., description="ARV variance
metric")
    discount_rate: float = Field(0.05, description="Systemic r metric
discount factor")
    projection_years: int = Field(50, description="Lifecycle evaluation
timeframe")

class JudicialClaimInput(BaseModel):
    claim_type: str = Field(..., description="PATENT / TRADE_SECRET /
BREACH_OF_CONFIDENCE / STATUTORY_MISCONDUCT")
    allegation: str = Field(..., description="Detailed description of the
claim cause")
    evidence_preservation_risk: bool = Field(True, description="Indicates
active risk of data loss or alteration")
    quantifiable_benefit_cad: float = Field(..., description="Avoided costs
or attributable revenue pool")

# =====
# 4. FASTAPI API BACKEND IMPLEMENTATION
# =====
app = FastAPI(
    title="DWVSCPS MASTERIZED ENTERPRISE TRIPLE THREAT SCROLL API",
    version="5.0.0",
    description="Unified verification engine bridging technical IP telemetry,
tax credit algorithms, and judicial bundles."
)

@app.get("/health")
def health_check():

```

```

    return {
        "status": "OPERATIONAL",
        "timestamp": utc_now(),
        "system_id": MASTER_CONFIG["system_id"],
        "master_hash_anchor":
MASTER_CONFIG["cryptographic_anchors"]["master_hash_anchor"]
    }

@app.get("/profile/ownership")
def get_ownership_profile():
    """Returns the absolute hardcoded corporate ownership boundaries and
    legal rights."""
    return {
        "owner_profile": MASTER_CONFIG["owner"],
        "identifiers": MASTER_CONFIG["administrative_registry"],
        "portfolio": MASTER_CONFIG["ip_portfolio_assets"],
        "legal_framework": MASTER_CONFIG["judicial_framework"]
    }

@app.post("/triple-threat/proof-1-technical")
def evaluate_technical_proof(payload: TelemetryDataInput):
    """Executes the core fluid velocity and SMT3 capture efficiency formulas
    (Proof Matrix 1)."""
    try:
        #  $V = C_d \cdot \sqrt{(2 \cdot \Delta P) / \rho}$ 
        inside_sqrt = (2.0 * payload.pressure_differential_pa) /
payload.fluid_density_kg_m3
        velocity = payload.discharge_coefficient * (inside_sqrt ** 0.5) if
inside_sqrt >= 0 else 0.0

        # Approximate  $\hat{I}$  capture metric step mapping
        conc_diff = payload.gas_concentration_in -
payload.gas_concentration_out
        capture_efficiency = (conc_diff / payload.spatial_containment_phi) *
payload.thermodynamic_scalar_psi if payload.spatial_containment_phi != 0 else
0.0

        result_id = f"PRF1-{uuid.uuid4().hex[:12].upper()}"
        result = {
            "proof_id": result_id,
            "timestamp": utc_now(),
            "metrics": {
                "calculated_bypass_velocity_m_s": velocity,
                "calculated_capture_efficiency_scalar": capture_efficiency
            },
            "status": "VERIFIED_TEMPORAL_BASELINE"
        }

        log_audit("EVALUATE_TECHNICAL_PROOF", result_id, "SUCCESS")
        return result
    except Exception as e:
        raise HTTPException(status_code=400, detail=str(e))

@app.post("/triple-threat/proof-2-tax")
def evaluate_tax_proof(payload: TaxAuditInput):

```

```

    """Computes the 50-year parallel asset Net Tax Efficiency algorithm
    (Proof Matrix 2)."""
    try:
        nte_sum = 0.0
        years = payload.projection_years if payload.projection_years > 0 else
50
        for t in range(1, years + 1):
            numerator = (payload.annual_operational_savings +
                payload.carbon_credit_indexing) - (payload.verification_monitoring_cost *
                payload.audit_variance_coefficient)
            denominator = (1.0 + payload.discount_rate) ** t
            nte_sum += numerator / denominator

        result_id = f"PRF2-{uuid.uuid4().hex[:12].upper()}"
        result = {
            "proof_id": result_id,
            "timestamp": utc_now(),
            "evaluation": {
                "net_tax_efficiency_yield_cad": nte_sum,
                "timeframe_years": years,
                "discount_factor_applied": payload.discount_rate
            },
            "registry_alignment":
MASTER_CONFIG["administrative_registry"]["cra_personal_information_banks"],
            "status": "DETERMINED_COMPLIANCE_DATASET"
        }

        log_audit("EVALUATE_TAX_PROOF", result_id, "SUCCESS")
        return result
    except Exception as e:
        raise HTTPException(status_code=400, detail=str(e))

@app.post("/triple-threat/proof-3-judicial")
def register_judicial_claim(payload: JudicialClaimInput):
    """Maps claims against structural safeguards to compile courtroom bundles
    (Proof Matrix 3)."""
    claim_id = f"CLM-{uuid.uuid4().hex[:12].upper()}"

    required_proofs = ["established_liability", "causal_connection",
        "quantifiable_quantum"]
    if payload.evidence_preservation_risk:
        required_proofs.append("risk_of_immediate_alteration")

    record = {
        "claim_id": claim_id,
        "registered_at": utc_now(),
        "input_parameters": payload.model_dump(),
        "safeguards": {
            "jurisdiction_target":
MASTER_CONFIG["judicial_framework"]["court"],
            "judicial_centre":
MASTER_CONFIG["judicial_framework"]["judicial_centre"],
            "procedural_pathway":
MASTER_CONFIG["judicial_framework"]["summary_procedure"],
            "remedy_proof_requirements": required_proofs
        }
    }

```

```

        },
        "legal_effect": "ISSUE_REQUIRING_PROOF_AND_COMPLIANCE_REVIEW"
    }

    court_records = read_json(COURT_DB, [])
    court_records.append(record)
    write_json(COURT_DB, court_records)

    log_audit("REGISTER_JUDICIAL_CLAIM", claim_id, "FILED")
    return record

@app.get("/triple-threat/export-master-scroll")
def api_export_master_scroll():
    """Generates, seals, and signs an unassailable data-minimization master
    JSON package."""
    try:
        package = {
            "master_metadata": {
                "system_id": MASTER_CONFIG["system_id"],
                "version": MASTER_CONFIG["version"],
                "classification": MASTER_CONFIG["classification"],
                "exported_at": utc_now()
            },
            "profile_ownership": MASTER_CONFIG["owner"],
            "registries": MASTER_CONFIG["administrative_registry"],
            "cryptographic_seals": MASTER_CONFIG["cryptographic_anchors"],
            "judicial_ledger": read_json(COURT_DB, []),
            "audit_logs": read_json(AUDIT_DB, []),
            "legal_status":
"STRUCTURED_MEGA_SCROLL_EVIDENCE_PACKAGE_NOT_A_JUDICIAL_ORDER"
        }

        # Compute dynamic validation signature block
        raw_bytes = json.dumps(package, sort_keys=True).encode('utf-8')
        dynamic_signature = hmac.new(
MASTER_CONFIG["cryptographic_anchors"]["master_hash_anchor"].encode('utf-8'),
            raw_bytes,
            hashlib.sha256
        ).hexdigest()

        package["master_metadata"]["hmac_sha256_verification_seal"] =
dynamic_signature

        filename = ROOT /
f"exports/DWVSCPS_MASTER_SCROLL_{datetime.now(timezone.utc).strftime('%Y%m%dT%H%M%S')}.json"
        write_json(filename, package)

        log_audit("EXPORT_MASTER_SCROLL", str(filename), "SUCCESS")
        return {"status": "SUCCESS", "sealed_evidence_file": str(filename),
"verification_signature": dynamic_signature}
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))

if __name__ == "__main__":

```

```
    print(f"=== {MASTER_CONFIG['system_id']} INDEPENDENT WORKLOAD ENGINE  
===")  
    print(f"[â€œ] Immutability Anchor:  
{MASTER_CONFIG['cryptographic_anchors']['master_hash_anchor']}")  
    print(f"[â€œ] Fast API script initialized for independent sandbox  
compilation.")
```